

Manuscript Title: Four-Mode Electrochromic Windows Integrated with Energy Storage
Enabled by MnWO_4

Dear Editor,

We are pleased to submit our manuscript entitled "*Four-Mode Electrochromic Windows Integrated with Energy Storage Enabled by MnWO_4* " for your consideration.

In this work, we present a comprehensive strategy to

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We believe our work delivers a compelling combination of electrochemical performance optimization and interfacial mechanical insight, contributing a valuable perspective to the field of metal-anode stabilization. We kindly ask that our manuscript be considered for publication in your esteemed journal.

Thank you for your time and consideration.

Sincerely,
Seung-Mo Lee, Prof. Dr.

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Four-Mode Electrochromic Windows Integrated with Energy Storage Enabled by MnWO_4

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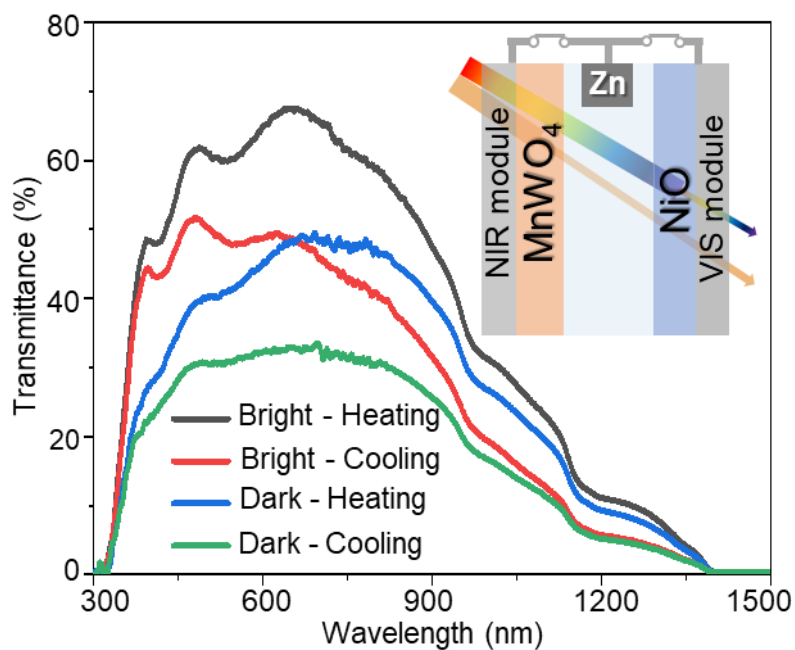
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TOC graphic and summary:

“ MnWO_4 serves as a multifunctional NIR electrochromic and energy-storage electrode, enabling reversible optical switching and charge storage in a four-mode electrochromic window.”



Abstract

Dual-band electrochromic windows that independently regulate visible (VIS) and near-infrared (NIR) solar radiation offer an attractive route to building energy management, yet most reported devices access only three optical states and rarely integrate energy storage. Here we introduce MnWO_4 as a dual-function electrode delivering both selective NIR electrochromism and charge storage. In a $\text{MnWO}_4\|\text{Zn}\|\text{NiO}$ architecture, a shared Zn anode allows independent addressing of the NIR-active MnWO_4 and VIS-active NiO electrodes, unlocking all four operational modes: bright-warm, bright-cool, dark-warm, and dark-cool. The NIR response originates from localized surface plasmon resonance, while storage arises from Zn^{2+} intercalation into the tungstate framework combined with a Mn-centered conversion reaction, giving high areal capacity over 200 cycles. Under simulated sunlight, surface temperatures differ by more than 6 °C between the bright-warm and dark-cool states, establishing an integrated platform for season-adaptive spectral control with on-board energy storage.

Keywords: smart window, electrochromic device, energy storage, MnWO_4 , four modes

Main Text

Electrochromic (EC) smart windows, which dynamically regulate solar radiation under applied bias, have emerged as a key technology for reducing building energy consumption associated with lighting, heating, and cooling (1–4). As buildings account for a substantial fraction of global energy use and carbon emissions, controlling solar heat gain and visible light transmission through windows offers a direct pathway toward improved energy efficiency (2, 4). Solar irradiation primarily consists of visible (VIS, 380–780 nm) and near-infrared (NIR, 780–2500 nm) components, where VIS light governs daylighting and visual comfort, while NIR radiation is responsible for heat transfer (5). Consequently, dual-band electrochromic systems capable of independently modulating VIS and NIR transmittance have attracted increasing attention for all-season energy management.

Despite notable progress, most reported dual-band EC devices remain limited in functionality. Conventional systems typically rely on a single electrochromic material or coupled redox processes, leading to synchronized modulation of VIS and NIR regions and restricting operation to three optical states: bright (VIS/NIR transparent), cool (VIS transparent/NIR blocked), and dark (VIS/NIR blocked) (2, 6–11). However, a fully adaptive smart window requires four distinct modes, including an additional “warm” state (VIS blocked/NIR transparent), enabling independent control over lighting and thermal radiation (12–14). Achieving such four-mode operation remains challenging due to the difficulty in decoupling VIS and NIR responses while maintaining electrochemical compatibility and device stability (13).

Another emerging direction in electrochromic research is the integration of energy storage functionality into EC platforms (15–17). Electrochromic energy-storage devices, including electrochromic batteries and supercapacitors, offer the dual advantage of modulating optical properties while storing and delivering electrical energy. This integration is particularly attractive for self-powered smart windows, where harvested or externally supplied energy can be stored and reused, thereby enhancing overall system efficiency. However, the combination

of materials and device architectures that simultaneously enable dual-band optical control and efficient energy storage remains well developed.

From a materials perspective, transition metal oxides have dominated EC research, particularly for NIR modulation via localized surface plasmon resonance or polaronic absorption (18). In contrast, transition metal tungstates remain relatively unexplored despite their rich redox chemistry and structural diversity. Among them, MnWO_4 is a promising candidate due to the coexistence of Mn redox centers and a tungsten-based framework, which can facilitate ion intercalation, electron transport, and charge compensation (19, 20). The multivalent nature of manganese ($\text{Mn}^{2+}/\text{Mn}^{3+}$) enables reversible Faradaic reactions that are not only beneficial for electrochromic modulation, especially in the NIR region, but also contribute to pseudocapacitive energy storage. This dual functionality makes MnWO_4 particularly attractive for coupling electrochromism with energy storage within a single material platform. Meanwhile, NiO is a well-established anodic electrochromic material exhibiting stable and reversible coloration in the VIS region, making it an ideal counterpart for selective visible-light modulation.

Herein, we report a dual-band electrochromic energy-storage device that integrates MnWO_4 and NiO electrodes to achieve independent regulation of NIR and VIS light. By combining MnWO_4 as the NIR-active electrode and NiO as the VIS-active electrode within a unified electrochemical system, the device enables four distinct optical modes: bright (VIS/NIR transparent), cool (NIR blocked/VIS transparent), warm (VIS blocked/NIR transparent), and dark (VIS/NIR blocked). Importantly, the redox processes governing electrochromism also contribute to charge storage, allowing the device to function as an energy-storage unit. These findings establish MnWO_4 as a promising multifunctional electrode material and highlight its potential for integration into next-generation multifunctional smart window systems that combine energy storage, electrochromic regulation, and enhanced building energy efficiency.

Design Principles of the Four-Mode Electrochromic Device

The design concept of the four-mode electrochromic window is illustrated in Fig. 1. The device adopts a three-electrode architecture in which a MnWO_4 NIR module and a NiO VIS module are independently addressable through a shared Zn anode. This configuration enables decoupled electrochemical control of the two complementary electrodes, which is the key prerequisite for achieving four distinct operational modes. In the bright–warm mode, both electrodes are in the bleached state, allowing high transmittance of both VIS and NIR radiation simultaneously (Fig. 1a). This mode is suitable for cold seasons or low-light conditions where maximum solar heat gain and indoor illumination are desirable. In the bright–cool mode, the NiO VIS module is colored while MnWO_4 remains bleached, selectively blocking visible light while permitting NIR transmission (Fig. 1b). In the dark–warm mode, MnWO_4 is colored to block NIR radiation while NiO remains bleached, preserving daylight transmission while reducing solar heat gain (Fig. 1c). This mode is particularly valuable in hot seasons where thermal control is needed without sacrificing natural lighting. Finally, in the dark–cool mode, both electrodes are colored, resulting in simultaneous blocking of VIS and NIR radiation for maximum solar rejection (Fig. 1d). The realization of all four modes, particularly the elusive dark–warm state, distinguishes this device from most reported dual-band electrochromic windows, which are typically limited to three optical states. The independent control of VIS and NIR transmittance enabled by the three-electrode Zn-anode architecture is thus central to the versatility and season-adaptive capability of the device.

Pristine WO_3 adopts a classical three-dimensional framework of corner-sharing WO_6 octahedra (Fig. 1e). While this sustains reliable electrochromic modulation, the specific capacitance remains inherently low: charge storage in standalone WO_3 is confined by bulk ion diffusion and restricted to the available crystallographic voids, preventing it from serving as an active pseudocapacitive matrix (21). MnWO_4 , in contrast, adopts a monoclinic wolframite chain configuration of edge-sharing MnO_6 and WO_6 octahedra (Fig. 1f). This open arrangement exposes abundant surface active sites to the electrolyte, and the multiple Mn oxidation states enable fast, reversible Faradaic redox at the surface and near-surface region. We believe that

MnWO₄ combines excellent electrochromism with robust pseudocapacitive storage, marking it as a true bifunctional smart-window electrode.

Characterization of MnWO₄

MnWO₄ was synthesized using a facile hydrothermal method (Fig. S1). Its structural, morphological, and chemical properties were systematically characterized. Scanning electron microscopy (SEM) imaging revealed a nanowire-like morphology with densely interconnected features distributed uniformly across the film surface (Fig. 2a). This porous, high-surface-area microstructure is expected to facilitate electrolyte penetration and ion transport during both electrochromic switching and energy storage operation. Low-magnification transmission electron microscopy (TEM) analysis further confirmed the nanowire-like morphology (Fig. 2b). The high-resolution TEM image exhibited clear lattice fringes with an interplanar spacing of 3.75 Å, corresponding to the (110) crystal plane of MnWO₄ (Fig. 2c) (22). This assignment was further corroborated by the selected area electron diffraction (SAED) pattern (Fig. S2). The X-ray diffraction (XRD) pattern confirmed the phase purity of the as-synthesized MnWO₄, with all diffraction peaks in good agreement with the monoclinic wolframite crystal structure (Fig. 2d and Fig.S3). No secondary phases were detected, indicating a clean and well-controlled synthesis. X-ray photoelectron spectroscopy (XPS) was performed to examine the valence states of tungsten and manganese. The high-resolution W 4f spectrum exhibited two sets of peaks assigned to W⁶⁺ 4f_{5/2} and W⁶⁺ 4f_{7/2}, confirming that tungsten is predominantly in the +6 oxidation state (Fig. 2e). The Mn 2p spectrum showed characteristic peaks corresponding to Mn²⁺ 2p_{1/2} and Mn²⁺ 2p_{3/2}, accompanied by satellite features consistent with high-spin divalent manganese in an octahedral coordination environment (Fig. 2f) (23). These results collectively confirm the successful formation of stoichiometric, phase-pure MnWO₄ with the expected electronic structure, providing a well-defined starting point for investigating its electrochromic and energy storage properties.

Energy Storage Performance of MnWO₄

The electrochemical energy storage behavior of MnWO₄ was evaluated in a Zn||MnWO₄ cell. The CV curve recorded at a scan rate of 0.1 mV s⁻¹ revealed multiple distinct redox features, indicating a complex, multi-step electrochemical process (Fig. 3a). Two cathodic reactions were identified upon discharge. At moderate potentials, Zn²⁺ ions intercalate into the WO₆ octahedra of MnWO₄ to form Zn_xWO₃, contributing to capacity through an intercalation mechanism. At lower potentials, a conversion reaction occurs in which MnO₆ octahedra is reduced by Zn²⁺ to form Zn_xMnO. The corresponding anodic process upon charging involves the decomposition of Zn_xMnO back to MnO, confirming the reversibility of the conversion reaction. The coexistence of intercalation and conversion pathways within a single electrode material is notable, as it enables MnWO₄ to deliver higher capacity than materials relying on a single storage mechanism alone. In contrast, the CV curve of commercial WO₃ exhibited only one broad peak (Fig. S4), indicating limited redox chemistry of WO₃. Galvanostatic charge-discharge profiles measured at current densities ranging from 0.01 to 0.1 mA cm⁻² demonstrated well-defined plateaus consistent with the redox processes identified in the CV (Fig. 3b). The cell delivered an areal capacity of approximately 2 mAh cm⁻² at the lowest current density of 0.01 mA cm⁻², reflecting the high electrochemical utilization of the MnWO₄ electrode. Rate capability testing showed a gradual decrease in capacity with increasing current density, retaining a meaningful capacity at 0.1 mA cm⁻², which is indicative of reasonably fast ion transport kinetics (Fig. 3c). Long-term cycling stability was assessed at both 0.02 and 0.1 mA cm⁻² over 200 consecutive cycles (Fig. 3d–e). At both current densities, the areal capacity remained stable throughout cycling with coulombic efficiency maintained close to 100%, demonstrating that the MnWO₄ electrode undergoes highly reversible electrochemical reactions without significant capacity fade. These results establish MnWO₄ as a capable energy storage electrode material, complementing its electrochromic functionality within the integrated device.

Electrochromic Performance of MnWO₄

The electrochromic properties of MnWO₄ were characterized in the Zn||MnWO₄ cell configuration. Digital photographs captured at different applied voltages clearly demonstrated a reversible and pronounced color transition (Fig. 4a). At 1.2 V, the electrode appeared in a light, near-transparent bleached state. Upon reducing the voltage to 0.0 V, the electrode darkened significantly into a deep blue-grey colored state. The original bleached appearance was fully recovered upon reapplying 1.2 V, confirming excellent optical reversibility. Transmittance spectra measured across the 300–1500 nm range at 0.0, 0.5, and 1.2 V revealed that the coloration effect was most pronounced in the NIR spectral region, with a large transmittance contrast between the colored and bleached states (Fig. 4b). The VIS transmittance, while also modulated, changed to a lesser extent relative to the NIR range. This spectral selectivity toward NIR modulation is consistent with an LSPR-driven electrochromic mechanism, in which free carrier absorption at low potentials preferentially attenuates NIR radiation before the onset of deeper polaronic coloration in the VIS range. The intermediate transmittance profile at 0.5 V further illustrates the gradual and voltage-tunable nature of the NIR modulation. The long-term optical stability of the MnWO₄ electrode was assessed by comparing transmittance spectra before and after 500 coloration-bleaching cycles (Fig. 4c). The spectra at both the colored and bleached states after 500 cycles closely overlapped with those recorded at cycle 1, demonstrating negligible degradation in optical contrast and spectral character. This electrochromic durability, combined with the energy storage stability demonstrated in Fig. 3, underscores the robustness of MnWO₄ as a dual-function electrode material.

Four-Mode Electrochromic Device Performance

The complete MnWO₄||Zn||NiO three-electrode device was assembled, and its four-mode optical operation was systematically characterized. The MnWO₄ electrode is responsible for NIR regulation, while NiO electrode serves as VIS module (Fig. S5–S7). It should be noted

that NiO exhibits limited stability. Therefore, improving its cycling stability should be a key focus of future work. Digital photographs of the device in each of the four modes showed clearly distinguishable optical appearances, confirming that each mode produces a visually distinct state (Fig. 5a). The bright–warm mode appeared the lightest and most transparent, while the dark–cool mode was the darkest. The bright–cool and dark–warm modes exhibited intermediate and spectrally differentiated appearances, reflecting the selective coloration of either the NiO VIS module or the MnWO₄ NIR module, respectively. Broadband transmittance spectra across the 300–1500 nm range provided quantitative confirmation of the spectral selectivity achieved in each mode (Fig. 5b). The bright–warm state exhibited the highest transmittance across both the VIS and NIR bands. In the bright–cool state, transmittance in the VIS region was substantially reduced while NIR transmittance remained comparatively high, consistent with selective NiO coloration. In the dark–warm state, NIR transmittance was preferentially attenuated by MnWO₄ coloration while VIS transmittance was better preserved, demonstrating the sought-after spectral inversion relative to the bright–cool mode. The dark–cool state showed the lowest transmittance across both bands, corresponding to simultaneous coloration of both electrodes. The ability to independently switch between these four spectrally distinct states within a single device represents a significant advance over conventional three-mode dual-band EC windows. The practical thermal management utility of the device was validated through solar simulator experiments monitored by infrared thermal imaging (Fig. 5c). A black heat-absorbing textile placed beneath the device was illuminated for up to 600 s. For comparison, under normal glass, the surface temperature increased rapidly, reaching 42.1 °C after 600 s of illumination (Fig. S8). In the bright–warm mode, the textile temperature rose from an initial 24.5 °C to 32.1 °C after 10 s, 34.4 °C after 60 s, and 37.2 °C after 600 s of continuous illumination (Fig. 5d). In contrast, the dark–cool mode markedly suppressed heat accumulation under identical conditions, with the temperature reaching only 29.2 °C after 10 s, 30.7 °C after 60 s, and 31.1 °C after 600 s. Notably, after 600 s, the temperature beneath the electrochromic device in the dark–cool mode was approximately 11 °C lower than that

under normal glass and 6 °C lower than that in the bright–warm mode. These results clearly demonstrate the device's effectiveness in regulating solar heat gain and reducing thermal load. These results highlight the potential of the $\text{MnWO}_4\|\text{Zn}\|\text{NiO}$ device as a practical, multifunctional smart window that combines season-adaptive solar regulation with integrated electrochemical energy storage.

In summary, we demonstrated a four-mode dual-band electrochromic window with integrated energy storage based on a $\text{MnWO}_4\|\text{Zn}\|\text{NiO}$ three-electrode architecture. MnWO_4 was introduced for the first time as a dual-function electrode material, delivering selective NIR electrochromic modulation through an LSPR-driven mechanism alongside high areal capacity through combined Zn^{2+} intercalation and conversion reactions. The independent electrochemical addressability of the MnWO_4 NIR module and NiO VIS module enabled full four-mode operation, bright–warm, bright–cool, dark–warm, and dark–cool, including the elusive dark–warm state that remains inaccessible in most reported dual-band devices. Infrared thermal imaging confirmed a surface temperature difference of over 6 °C between the bright–warm and dark–cool modes under simulated solar illumination, validating the practical thermal regulation capability of the device. The integration of season-adaptive spectral control and on-board energy storage within a single platform represents a meaningful step toward self-sustaining, energy-efficient building envelope technologies.

Acknowledgments

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Conflict of Interest

The authors declare no conflicts of interest.

Supporting Information

The Supporting Information is available at ...

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Figures

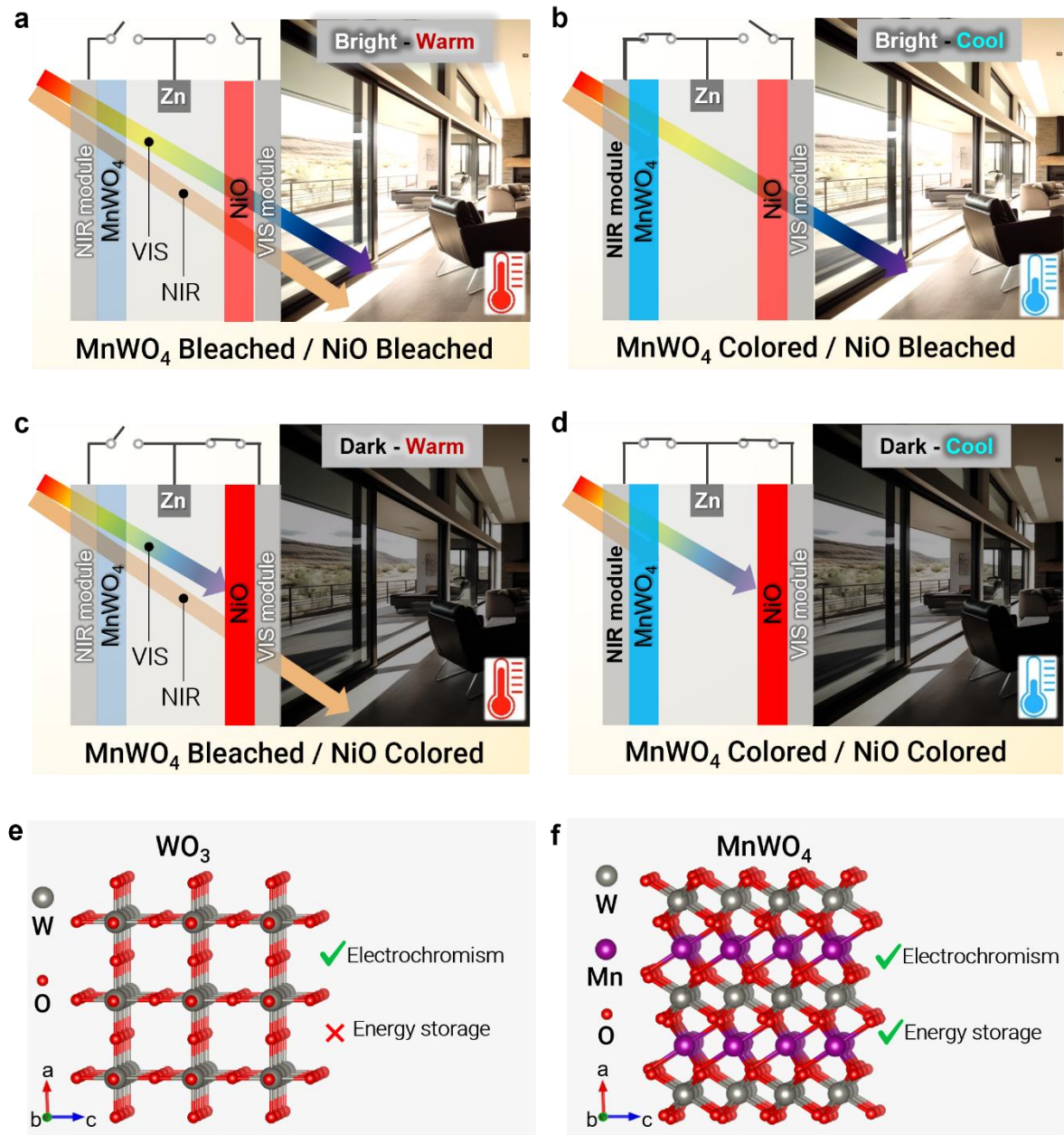


Figure 1. Design principles of the four-mode electrochromic windows. a) Bright-warm mode which allows high transmittance of both VIS and NIR ranges. **b)** Bright-cool mode which allows high transmittance of NIR range while blocking the VIS range. **c)** Dark-warm mode which allows high transmittance of VIS range while blocking the NIR range. **d)** Dark-cool mode which blocks both VIS and NIR ranges. **e–f)** Crystal structure of (e) WO₃ and (f) MnWO₄.

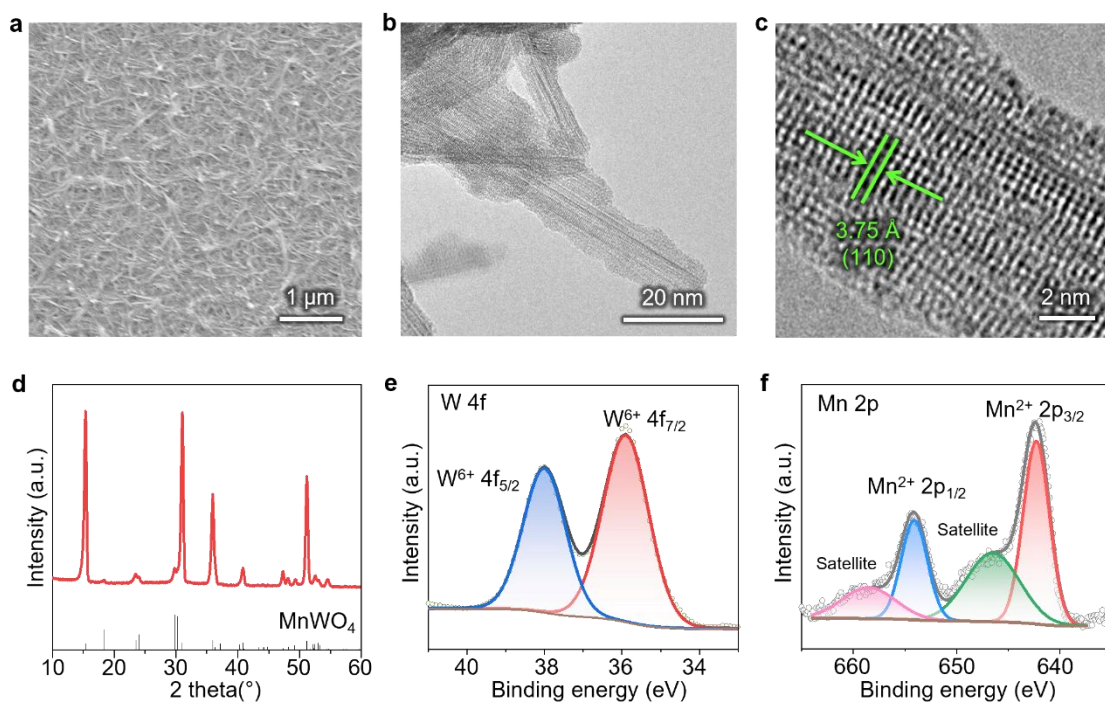


Figure 2. Characterization of MnWO₄. **a)** SEM image. **b–c)** TEM images. **d)** XRD spectrum. **e–f)** High-resolution XPS spectra of (e) W 4f and (f) Mn 2p.

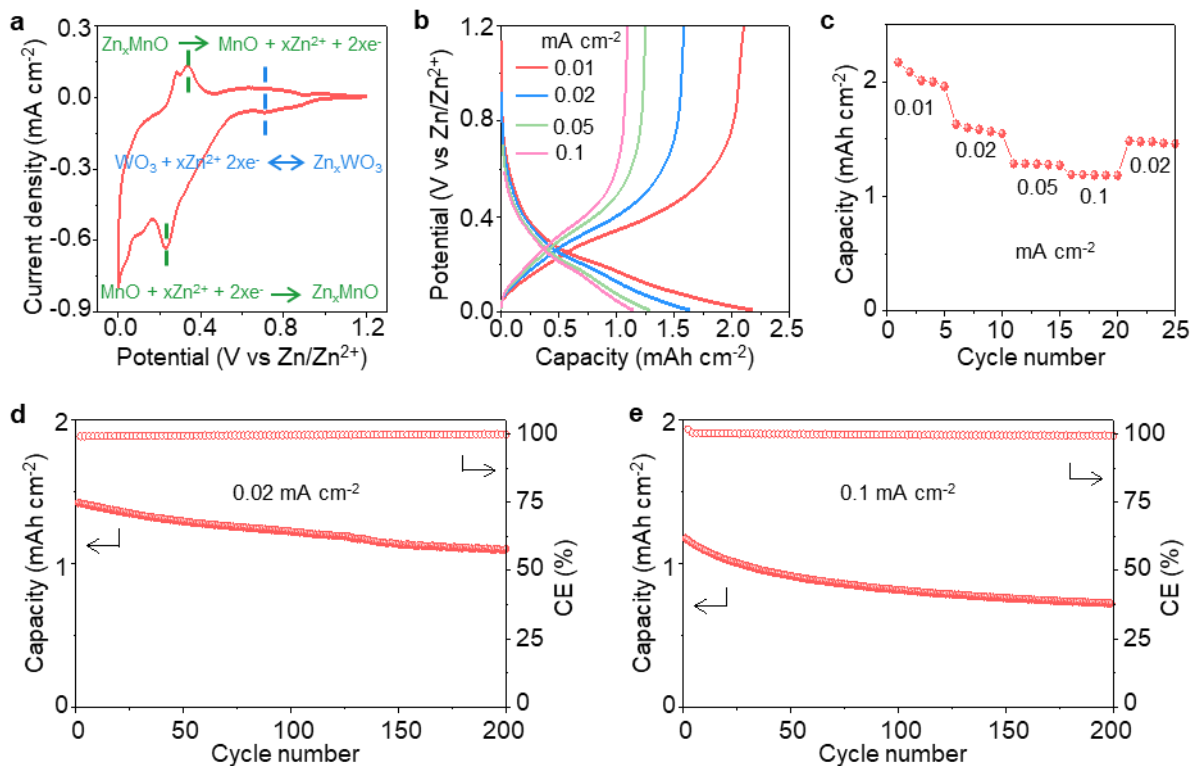


Figure 3. Energy storage performance of MnWO₄. **a)** CV curve at the scan rate of 0.1 mV s⁻¹. **b)** Charge-discharge curves at various current densities. **c)** Rate capability. **d–e)** Cycling performance at current densities of (d) 0.02 mA cm⁻² and (e) 0.1 mA cm⁻².

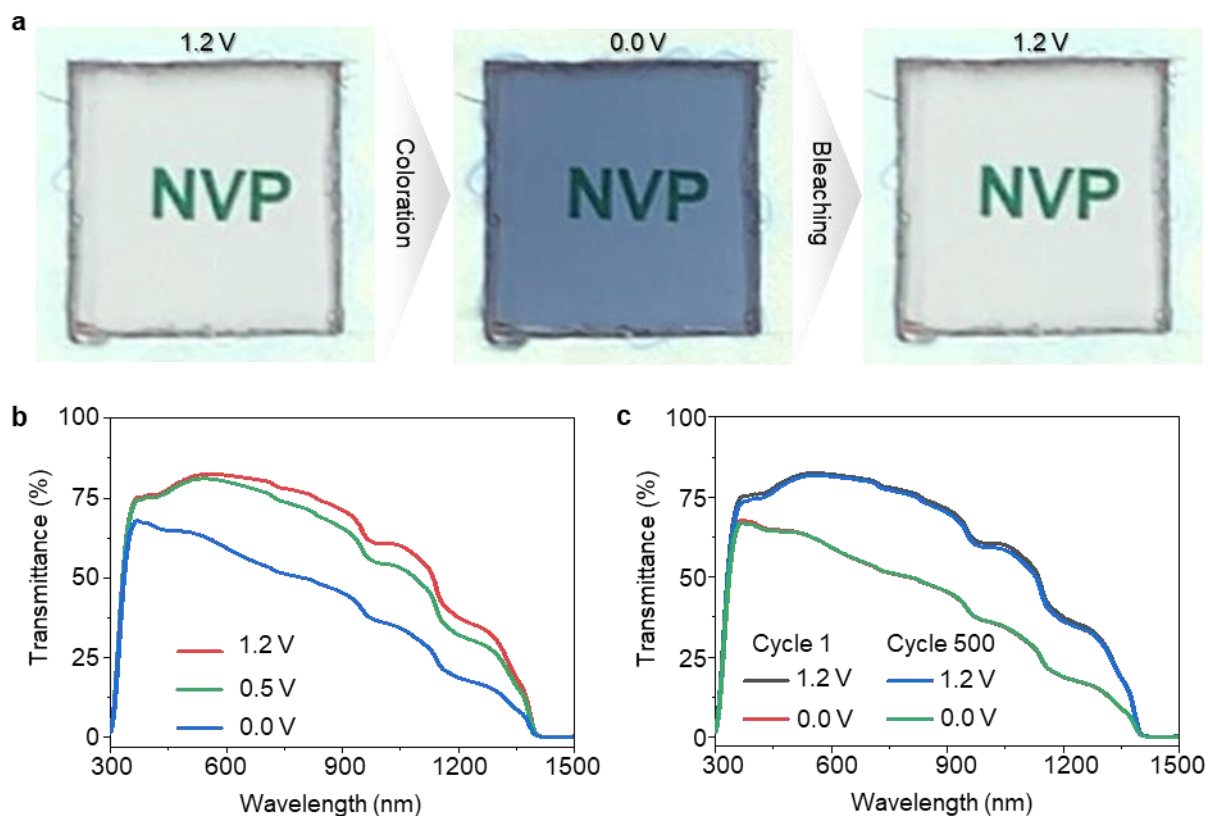


Figure 4. Electrochromic performance of MnWO_4 . **a)** Digital photos showing the color evolution of the $\text{Zn}||\text{MnWO}_4$ cell. **b)** Transmittance of the $\text{Zn}||\text{MnWO}_4$ cell at various coloration states. **c)** Transmittance of the $\text{Zn}||\text{MnWO}_4$ cell after 500 cycles of coloration and bleaching.

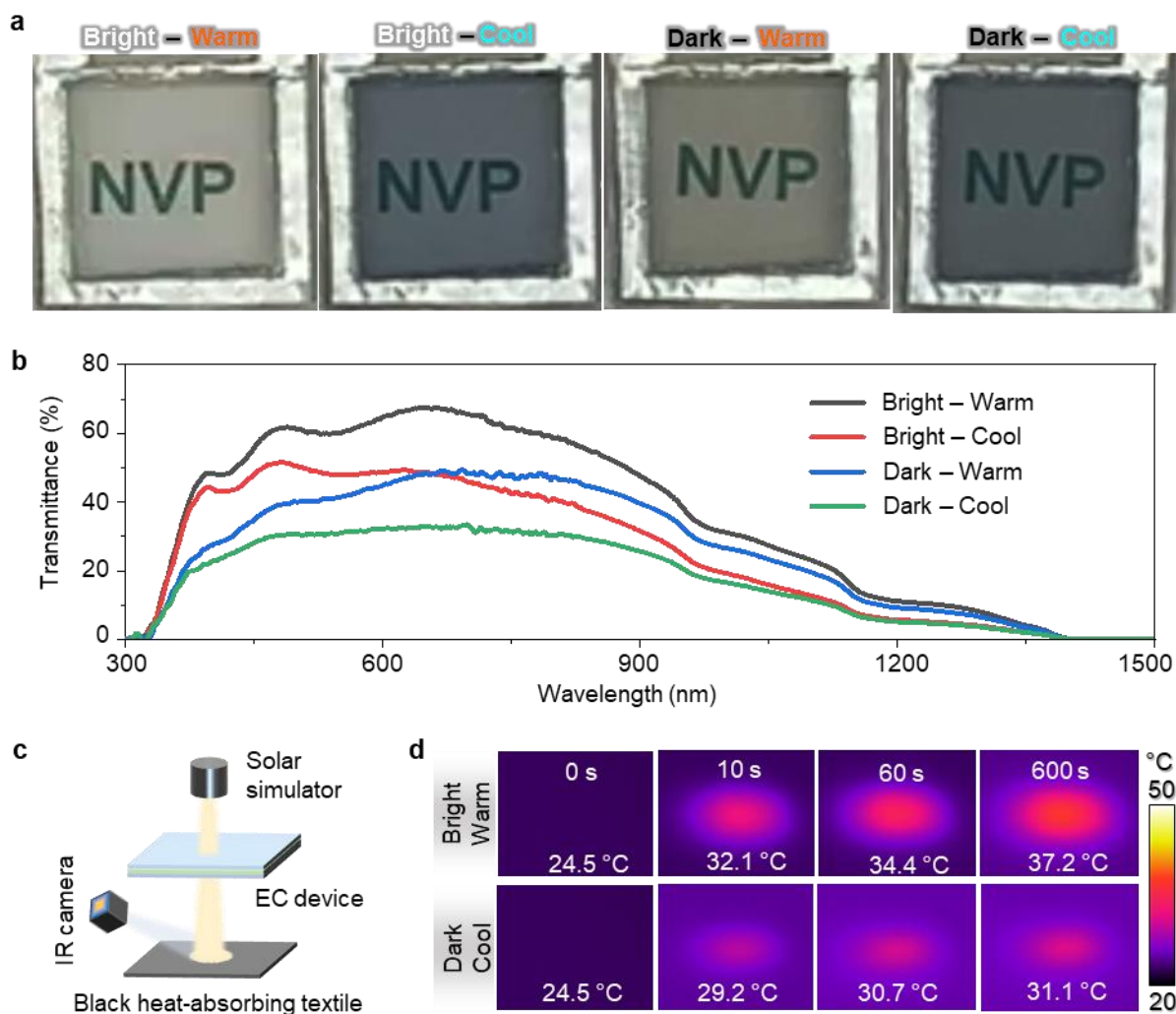


Figure 5. Four-mode electrochromic windows. **a)** Digital photos showing the color evolution of the $\text{MnWO}_4 \parallel \text{Zn} \parallel \text{NiO}$ cell at four different modes. **b)** Transmittance of the $\text{MnWO}_4 \parallel \text{Zn} \parallel \text{NiO}$ cell at four different modes. **c)** Setup **d)** Infrared thermal images of black heat-absorbing textile during being exposed to one solar simulator.

SUPPORTING INFORMATION

Four-Mode Electrochromic Windows Integrated with Energy Storage Enabled by MnWO₄

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1. Experimental Section

Preparation of MnWO₄-coated ITO glass

MnWO₄ was synthesized via a hydrothermal method. Briefly, 0.3 mmol of Na₂WO₄, 0.3 mmol of MnCl₂·4H₂O, and 0.1 mmol of Na₂SO₄ were dissolved in 40 mL of distilled water. The pH of the solution was adjusted to approximately 2 by the dropwise addition of HCl. The resulting solution was magnetically stirred for 1 h to ensure complete homogenization. Subsequently, the precursor solution was transferred into a 50 mL Teflon-lined stainless-steel autoclave. ITO glass substrates, pre-cleaned with ethanol and distilled water, were immersed in the solution inside the autoclave. The autoclave was sealed and heated at 180 °C for 1.5 h in a temperature-controlled oven, followed by natural cooling to room temperature. The resulting MnWO₄-coated ITO substrates were thoroughly rinsed with distilled water and ethanol and then dried under ambient conditions.

Preparation of NiO-coated ITO glass

The NiO on ITO glass was prepared by sputtering method. ...

Material characterizations

The surface morphology and elemental composition of the samples were characterized using field-emission scanning electron microscopy (FE-SEM, JSM-7800F, JEOL) equipped with an energy-dispersive X-ray spectroscopy (EDS) system (AZtec®, Oxford Instruments). Crystal structures were examined by X-ray diffraction (XRD) using an Empyrean diffractometer (PANalytical) with Cu K α radiation ($\lambda = 1.5418 \text{ \AA}$). The chemical composition and surface electronic states were investigated by X-ray photoelectron spectroscopy (XPS) using a MultiLab 2000 system (Thermo Fisher Scientific) equipped with an Al K α X-ray source. The microstructure of the MnWO₄ samples was further analyzed by transmission electron microscopy (TEM, Talos G2, Thermo Fisher Scientific) operated at an accelerating voltage of 200 kV.

Fabrication of electrochromic devices

The electrochromic devices were assembled in a sandwich configuration. For the half-cell configuration, a zinc foil frame was positioned between the MnWO₄- or NiO-coated ITO substrate and a glass cover. Specifically, a 1 mm-thick 3M double-sided adhesive tape with a centrally cut rectangular opening ($1.5 \times 1.5 \text{ cm}^2$) was attached to the MnWO₄- or NiO-coated ITO substrate, serving as both a spacer and sealing layer. A zinc foil frame pre-attached to a glass substrate was then aligned on top of the spacer to form a sealed cell. Finally, 0.5 M ZnSO₄ aqueous solution was injected into the cell as the electrolyte using a syringe. The full electrochromic devices were assembled using the same procedure with a zinc foil frame sandwiched between MnWO₄-coated ITO and NiO-coated ITO substrates.

Electrochemical measurements

All electrochemical measurements were performed using a potentiostat/galvanostat (VSP, Bio-Logic Science Instruments). The electrochemical performance of the electrochromic devices was evaluated in a two-electrode configuration, with the MnWO₄- or NiO-coated ITO

substrate serving as the working electrode and a transparent Zn mesh serving as both the counter and reference electrode. Optical characterization was conducted using a UV–Vis–NIR spectrophotometer (Shimadzu) over a wavelength range of 300–1500 nm. The optical transmittance of each electrode was measured using air as the reference (100% transmittance baseline).

2. Supporting Figures

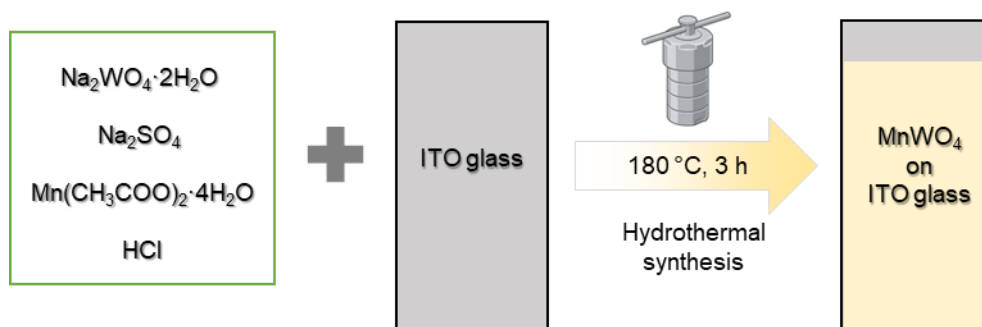


Figure S1. Schematic illustration showing the fabrication of MnWO₄.

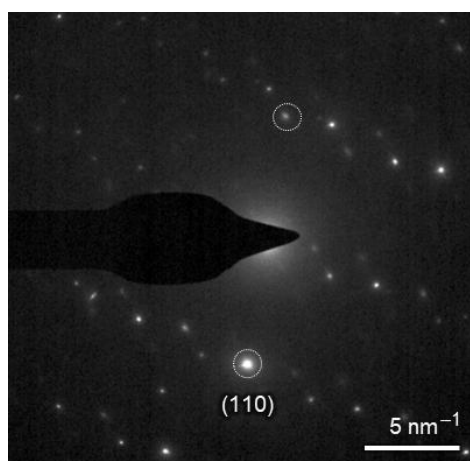


Figure S2. SAED pattern of MnWO₄.

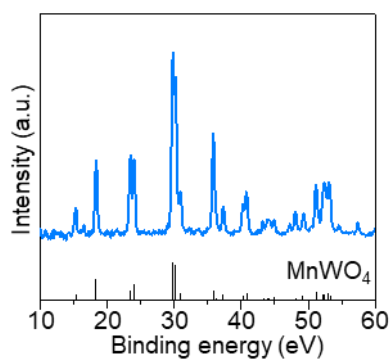


Figure S3. XRD pattern of MnWO₄ grown on the wall of the PTFE liner of autoclave reactor.

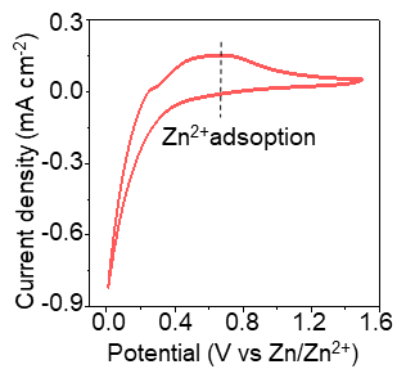


Figure S4. CV curve of the commercial WO_3 at the scan rate of 0.1 mV s^{-1} .



Figure S5. Digital photos showing the color evolution of the $\text{Zn}||\text{NiO}$ cell.

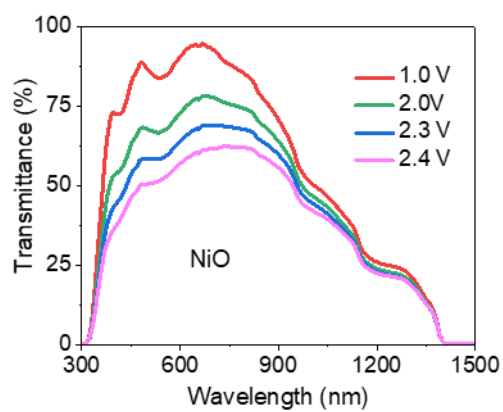


Figure S6. Transmittance of the $\text{Zn}||\text{NiO}$ cell at various coloration states.

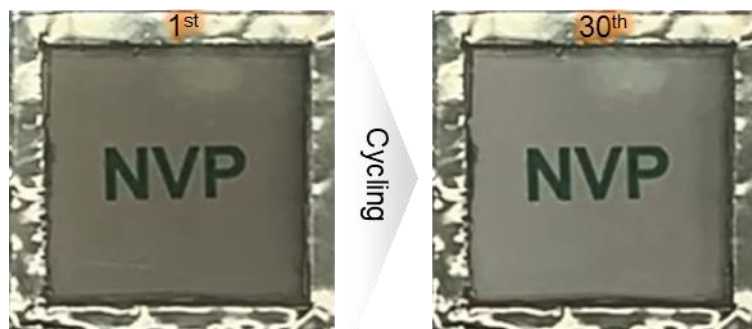


Figure S7. Digital photos showing the stability of the Zn||NiO cell after 30 cycles.

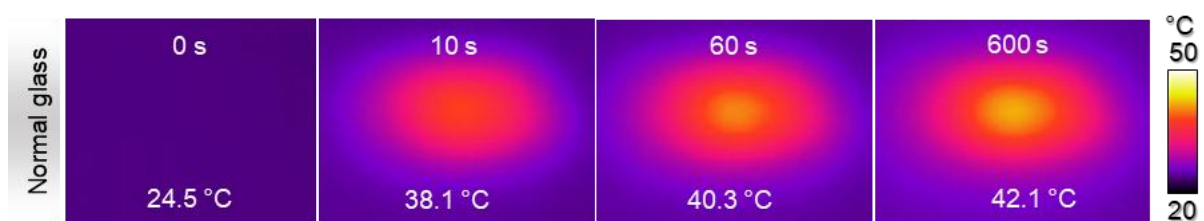


Figure S8. Infrared thermal images of black heat-absorbing textile during being exposed to one solar simulator under the normal glass.